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Unit – 1 : Session – 6 : SLO – 1 | 1's and 2's complement

■ MCQs

1. c — 2's complement
2. c — 01010 (invert all bits of 10101)
3. c — Flip the bits and add 1
4. d — Take 1's complement and mark result as negative
5. b — It is ignored (discarded)

■ Two-Mark Questions

1. 1's complement: invert every bit. Example: $1010 \rightarrow 0101$. 2's complement: invert every bit and add 1. Example: $1010 \rightarrow 0101 + 1 = 0110$.
2. Sign-magnitude: MSB is the sign, remaining bits store the magnitude; has +0 and -0; arithmetic is complex. 2's complement: negative number = 2's complement of its magnitude; single zero; addition/subtraction work with one simple adder — hence preferred.

■ Scenario-Based Long Answer

- 2's complement subtraction $A - B = A + (2\text{'s complement of } B)$. Example: $7 - 5$ with 4 bits: $A = 0111$, $B = 0101$. 2's complement of $B = 1010 + 1 = 1011$. Add: $0111 + 1011 = 1\ 0010$. A carry is generated $\rightarrow$ discard it $\rightarrow$ result $0010 = +2$ (positive).
- If NO carry is generated, the result is negative and is in 2's complement form; take its 2's complement to get the magnitude and attach a minus sign. Example: $5 - 7$: $0101 + 1001 = 1110$ (no carry) $\rightarrow$ 2's complement of $1110 = 0010 \rightarrow$ result = -2.

■ Match the Following

1. 1's Complement — b) Invert bits
2. 2's Complement — a) Add 1 to 1's complement
3. Sign-Magnitude — c) MSB for sign, rest for magnitude
4. No carry in 2's complement — d) Result is negative
5. Carry in 2's complement — e) Ignore carry, result is positive

■ Fill in the Blanks

1. 00110
2. 11011 ($01010+1$)
3. sign
4. negative (in 2's complement form)
5. end-around carry (the carry, i.e. 1, to LSB)

■ Scenario-Based Small Activity

- $A - B = 0110 - 0011$. 2's complement of B : $0011 \rightarrow 1100 + 1 = 1101$.
- Add: $0110 + 1101 = 1\ 0011$. Discard the carry $\rightarrow 0011$.
- Final result: $0011 = 3$ (positive, correct: $6 - 3 = 3$).

■ True or False

1. FALSE — They differ (2's complement = 1's complement + 1).
2. TRUE (end-around carry).
3. TRUE
4. FALSE — 2's complement = invert bits THEN add 1.
5. TRUE — discard carry, result is final.